

When Does Compressed KV Start Lying?

Token-Level Drift in KV Cache Compression

ExactKV Project
Technical Report (v2.9)

June 2026

Quantitative values from `reports/scale_7b/raw.json`, `reports/evidence_plus/raw.json`, and `reports/external_panels/summary_all.json`. Claim boundaries: `docs/CLAIM_BOUNDARIES.md`.

Abstract

Lossy KV-cache compression is widely deployed, but most evaluations report aggregate quality and stop there. ExactKV is a compressor-agnostic **crash-test framework** that measures first-divergence index, draft acceptance, and `exactkv_failure` under verifier-mediated semantics.

Across **7,348 completed GPU cells** (Llama-3.1-8B and Mistral-7B, five benchmark families) all panels report `exactkv_failures=0`. Four main findings:

1. **Task type dominates.** `int4_sim`: 6% (MBPP code) → 11% (BFCL short) → 50% (BFCL long-gen) → **90%** (HF LongBench). H2O-style eviction at 75% kept: **100%** divergence on LongBench.
2. **Generation length scales within-task.** On BFCL, `int4_sim` rises from 9% (mnt=16) to 62% (mnt=256) — a $7\times$ increase.
3. **Three distinct failure modes.** Top-k logit autopsy (1,103 divergent cells): near-tie noise (`int8`, rank 2.4, fdi=22); distribution shift (`int4_sim`, rank 3.5, fdi=8); attention destruction (H2O-style, rank 6.7, fdi=1). Three forensic case studies confirm each.
4. **100% downstream validity preserved.** Despite 50% token drift, ExactKV preserves all 106/106 full-KV valid BFCL tool calls (both models, all four task categories).

`int6_sim` (6-bit per-tensor) and `int4_per_vec_sim` (KIVI/KVQuant-style per-vector INT4, $2.06\times$ `int8` error) are GPU-validated non-catastrophic on Mistral-7B (v3.0, 784 cells, `exactkv_failures=0`): both show 0% divergence on BFCL and MBPP; `int6_sim` reaches 37.5% and `int4_per_vec_sim` 55.6% on HF LongBench — both non-catastrophic, with per-vector granularity task-conditionally dominating bit-width. Results are **not** benchmark scores, production claims, or a reproduction of VeriCache [8].

1 Introduction

Compressed KV is usually treated as a transparent optimization. Under greedy decoding, a single perturbed logit can flip the argmax, trajectories diverge. Aggregate metrics hide *where* divergence begins. ExactKV asks: **at what token does the compressed-cache path first stop matching full-KV greedy decoding?**

1.1 Shared problem framing with VeriCache

VeriCache [8] and ExactKV share the observation that lossy KV may look acceptable on short or aggregate metrics but **diverges more as decoding continues**, and catastrophic for code and tool-calling. VeriCache uses compressed KV to **draft** and full KV to **verify/correct**, guaranteeing **identical greedy output** to full-KV inference.

1.2 Where ExactKV diverges from VeriCache

VeriCache is a **serving/system** paper. Its contribution is not merely “draft then verify.” It makes that loop practical by: (1) keeping compressed KV on GPU, (2) keeping full KV in CPU/storage, loading only for verification, (3) **cross-resource staggering** (HBM-bound draft vs interconnect-bound verify), (4) reporting serving throughput (up to $\sim 4\times$ vs full-KV in their evaluation) with identical outputs.

ExactKV is an **evaluation crash-test**. It measures first divergence, acceptance, and exactness failures, *not* deployment throughput. It does *not* reproduce VeriCache scheduling or memory tiering.

1.3 Main Findings at a Glance

#	Finding	Key number
1	Task type dominates drift	int4_sim: MBPP 6% $\rightarrow$ BFCL 11% $\rightarrow$ BFCL long 50% $\rightarrow$ LongBench 90%
2	Generation length scales drift	int4_sim mnt=16: 9% $\rightarrow$ mnt=256: 62% ($7\times$)
3	Eviction > quantization	H2O-style 75% kept $\rightarrow$ 100% LongBench divergence
4	Three distinct failure modes	int8: near-tie (rank 2.4, fdi=22); int4_sim: shift (rank 3.5, fdi=8); H2O: destruction (rank 6.7, fdi=1)
5	100% downstream validity	ExactKV preserves 106/106 valid BFCL tool calls despite 50% drift
6	Zero correctness failures	exactkv_failures=0 across all 7,348 GPU cells

Table 1: Main findings summary.

2 Definitions

2.1 Reference paths

- **Full-KV reference:** greedy generation with uncompressed KV.
- **Lossy path:** compressed-KV greedy without verifier.
- **ExactKV path:** verifier-mediated draft/verify/commit (ExactKVGenerator).

2.2 Metrics (formal)

Corrections vs rejections: `total_rejected` counts rejected *draft tokens*, `total_corrections` counts verification *rounds* with a correction. For `int4_sim`: 1284/1502 accepted, 218 rejected tokens, **116 correction rounds**.

Table 2: Core metrics (formal definitions).

Metric	Definition
first_divergence_index	First position where the <i>lossy</i> token $\neq$ full-KV greedy (before correction).
acceptance_rate	$\text{total_accepted} / \text{total_drafted}$ across verification rounds.
exactkv_failure	Final ExactKV output $\neq$ full-KV greedy (verifier/commit failure).
divergence_rate	Fraction of cells with lossy-path divergence.
divergence_stability	$1 - \text{divergence_rate}$ (complement).
compression_ratio	Stored tensor byte ratio only (not active GPU memory).

2.3 Algorithm

```

Prefill -> FullKVState + CompressedKVState
while generated < max_new_tokens:
    DRAFT from compressed KV (up to draft_len tokens)
    VERIFY each token vs full-KV greedy
    COMMIT accepted prefix, correct on mismatch
    ALIGN compressed state from authoritative full KV
return ExactKV output_ids

```

Greedy decoding only in this release.

3 Method

Each **cell** is (model, compressor, prompt, max_new_tokens) with draft_len=4, seed=0. Three modes per cell: **full**, **lossy**, **exactkv**. Leaderboard score: $0.35a + 0.25v + 0.20(1 - \hat{d}) + 0.10(1 - f) + 0.10s$.

4 Experimental Setup

Table 3: Headline release panel (reports/scale_7b/raw.json).

Parameter	Value
Panel ID	phase_a_unified_panel
Total cells	1500 = $2 \times 5 \times 50 \times 3$
Models	Llama-3.1-8B (750), Mistral-7B (750)
Stack	torch 2.8.0+cu128, transformers 5.12.1
draft_len	4 max_new_tokens $\in \{4, 8, 16\}$
Decoding	Greedy only
Prompts	50 variants from 4 stress templates (capital, math, JSON, code)
Categories	capital_france 390, simple_math 390, json_tool 360, code_fn 360
Execution	Sequential per model, deterministic_mode=false

Validated compressors: noop, int8, int4_sim. Real KIVI/KVQuant/SnapKV/TurboQuant are not in the headline panel. The raw JSON also contains proxy/probe slots (**spectralquant**, **shard**) excluded from all analysis; see Limitations (§??).

1,560 GPU cells total: 216 Llama-3.1-8B cells across bundled Long-Bench/RULER/BFCL/HumanEval pilots; 144 MBPP cells across both models (mbpp_gpu_raw.json); 1,200 BFCL export-50 tool-call drift cells across both models — exactkv_failures=0 throughout. Not official benchmark scores.

5 Results: Validated Compressors

Table 4: built-in compressors with direct ExactKV integration (noop, int8, int4_sim). Aggregates from compressor_summary in reports/scale_7b/raw.json (300 cells each: 150 per model).

Compressor	Cells	Accept.	Div. [†]	Stab.	1st div. [§]	Fail.
noop	300	1.000	0.00	1.00	n/a	0.00
int8	300	1.000	0.00	1.00	n/a	0.00
int4_sim	300	0.851	0.52	0.48	2.0	0.00

Table 4: Built-in compressors from compressor_summary. [†]Lossy-path divergence fraction. [§]Mean over divergent cells only. Histogram for int4_sim: index 1→78, 3→39, 5→24, 8→13 cells.

int8/noop: no lossy divergence. int4_sim: drifts in 52% of cells yet exactkv_failure=0 because the verifier corrects.

Leaderboard-style scores (formula: $0.35a + 0.25v + 0.20(1 - \hat{d}) + 0.10(1 - f) + 0.10s$, source: reports/public_release/leaderboard_final.json):

Compressor	Model	Score	Accept.	Div. rate [†]	Stab.	Cells
noop	Llama-3.1-8B	1.000	1.000	0.000	1.000	150
int8	Llama-3.1-8B	1.000	1.000	0.000	1.000	150
noop	Mistral-7B	1.000	1.000	0.000	1.000	150
int8	Mistral-7B	0.983	1.000	0.000	0.827	150
int4_sim	Llama-3.1-8B	0.859	0.852	0.520	0.480	150
int4_sim	Mistral-7B	0.851	0.837	0.507	0.493	150

Table 5: Leaderboard-style scores for validated built-in compressors. [†]Div. rate = raw lossy-path token divergence fraction (same metric as Table 4). int8 Mistral: divergence_rate=0.000 (zero cells diverge); Stab.=0.827 is the **leaderboard stability subscore** from leaderboard_final.json, which is distinct from the formal metric divergence_stability = 1 − divergence_rate (Table 2); under the formal metric, int8 Mistral divergence_stability=1.000. spectralquant (MOCK→int4_sim) and shard (PROBE_ONLY) excluded from ranked table.

5.1 Metric interpretation

Divergence rate and acceptance rate measure *different* failure modes. Low lossy-path divergence does not imply high verifier acceptance (and vice versa). lossy-path divergence does not imply high verifier acceptance (and vice versa). Example: int4_sim at max_new_tokens=16 can show first divergence at index 8 with acceptance 0.94 — how late drift occurs matters.

5.2 Evidence-plus long-context panel

Source: reports/evidence_plus/raw.json (RunPod A5000, June 2026). 144 cells = $2 \times 6 \times 2 \times 3 \times 2$, prefill buckets 512/1024, max_new_tokens $\in \{16, 32\}$, built-in compressors only. By context

Compressor	Cells	Accept.	Div. rate	Fail.
noop	48	1.000	0.00	0.00
int8	48	1.000	0.00	0.00
int4_sim	48	0.985	0.31	0.00

Table 6: Evidence-plus aggregates. int4_sim: Mistral 41.7%, Llama 20.8% divergence (24 cells each).

bucket (all compressors): 512→11.1% divergence, 1024→9.7%. Diagnostic timing: mean 3854 ms, p90 5178 ms per cell (not serving throughput).

5.3 External benchmark smoke panels

Source: analysis_pack.json (216 Llama-only), mbpp_gpu_raw.json (144 cells). exactkv_failures=0 on all panels. 216-cell subset: noop/int8: 0%; 15 divergent cells (int4_sim only).

Panel	Source	Model	Ctx (K)	MNT	Cells	Div.	Acc.	Fail.	P90 ms
LongBench	pilot	Llama-3.1-8B	2,4	16,32	72	0.083	0.994	0	8781
RULER 2K-4K	pilot	Llama-3.1-8B	2,4	16,32	48	0.083	0.993	0	8793
RULER 8K	pilot	Llama-3.1-8B	8	16,32	24	0.083	0.993	0	13640
BFCL	pilot (4)	Llama-3.1-8B	1,2	16,32	48	0.063	0.999	0	6342
HumanEval	pilot (4)	Llama-3.1-8B	1,2	32	24	0.000	1.000	0	6392

Table 7: Llama-3.1-8B external smoke panels (216 cells, noop/int8/int4_sim). Drift metrics; not official scores.

Panel	Source	Models	Ctx (K)	MNT	Cells	Div.	Acc.	Fail.	P90 ms
MBPP	pilot (6)	Llama+Mistral	0.5,1	16,32	144	0.021	0.999	0	5121

Table 8: MBPP code-drift smoke (144 cells, noop/int8/int4_sim). Not pass@1; no test execution.

Notable: LongBench int4_sim 25%; RULER 8192 niah_single 33%; BFCL pilot int4_sim 18.75%; BFCL export-50 int4_sim 11.3% (Llama 5.5%, Mistral 17.0%); HumanEval zero drift; MBPP 3/144 divergent. int8/noop: 0% divergence on all panels.

Overall acceptance (initial 216 ext. pilot cells): Full-acceptance rate 93.7%, Wilson 95% CI [90.5%, 96.8%]. Source: analysis_pack.json → totals.acceptance_full_rate_ci95.

5.4 KIVI offline compressor panel — real HF results (640 cells)

Source: kivi_longbench_hf_raw.json (320 cells) + kivi_mbpp_hf_raw.json (320 cells). RunPod RTX A5000, June 2026. exactkv_failures=0 on all 640 cells.

Compressor	Model	Cells	Div. rate	CI ₉₅ low	CI ₉₅ high
noop	Llama-3.1-8B	200	0.000	0.000	0.019
noop	Mistral-7B	200	0.000	0.000	0.019
int8	Llama-3.1-8B	200	0.000	0.000	0.019
int8	Mistral-7B	200	0.000	0.000	0.019
int4_sim	Llama-3.1-8B	200	0.055	0.031	0.096
int4_sim	Mistral-7B	200	0.170	0.124	0.228

Table 9: BFCL export-50 tool-call drift panel (1,200 cells, both models, 50 prompts). `exactkv_failures=0`. Source: `bfcl_export_50_raw.json`. Drift measurement only (not JSON-completeness); `max_new_tokens` ∈ {16, 32}. Mistral-7B `int4_sim` divergence (17.0%) is 3× Llama-3.1-8B (5.5%).

Panel	n	Div.	Rate	CI ₉₅ low	CI ₉₅ high
LongBench pilot	72	6	8.3%	3.9%	17.0%
RULER 2K–4K	48	4	8.3%	3.3%	19.7%
RULER 8K	24	2	8.3%	2.3%	25.8%
BFCL pilot	48	3	6.3%	2.2%	16.8%
HumanEval pilot	24	0	0.0%	0.0%	14.2%
MBPP pilot	144	3	2.1%	0.7%	6.0%
BFCL exp-50 int4_sim Llama	200	11	5.5%	3.1%	9.6%
BFCL exp-50 int4_sim Mistral	200	34	17.0%	12.4%	22.8%
BFCL exp-50 int8/noop	800	0	0.0%	0.0%	0.5%
All 216 ext. cells	216	15	6.9%	4.3%	11.1%

Table 10: Wilson 95% CIs on divergence rate (panel-level). Source: `scripts/build_external_analysis_pack.py`. “All 216 ext. cells” refers to the **initial** LongBench/RULER/BFCL/HumanEval pilot panel (first workflow, Llama-only); the full 1,560-cell external smoke total includes MBPP and BFCL export-50.

Compressor	LB div.	LB acc.	MBPP div.	MBPP acc.	Fail.
noop	0.0%	1.000	0.0%	1.000	0
int8	13.8%	0.994	0.0%	1.000	0
int4_sim	91.2%	0.818	5.0%	0.995	0
kivi_offline	100.0%	0.004	100.0%	0.001	0

Table 11: KIVI offline compressor panel (80 cells each × 4 compressors × 2 datasets). `kivi_offline` produced catastrophic **adapter-level** drift in this ExactKV integration (near-zero acceptance). Because this is an offline simulate-path adapter rather than KIVI production CUDA/Triton serving, these results diagnose the adapter/integration path, *not* the KIVI algorithm as deployed. ExactKV detected and corrected all cells (`exactkv_failures=0`). Real HF `int4_sim` drift (LB: 91.2%) greatly exceeds bundled pilot (20.8%).

6 ExactKV-HF-LongBench Drift Panel (v2.6 — Complete)

Status: Complete. 720 cells, both models, `exactkv_failures=0`.

Model	Compressor	n	Div.	Rate	CI ₉₅	Mean acc.	EKV fail
Llama-3.1-8B	noop	120	0	0.0%	[0.0%, 3.1%]	1.000	0
Llama-3.1-8B	int8	120	33	27.5%	[20.3%, 36.1%]	0.988	0
Llama-3.1-8B	int4_sim	120	110	91.7%	[85.3%, 95.4%]	0.825	0
Mistral-7B	noop	120	0	0.0%	[0.0%, 3.1%]	1.000	0
Mistral-7B	int8	120	26	21.7%	[15.2%, 29.9%]	0.988	0
Mistral-7B	int4_sim	120	107	89.2%	[82.3%, 93.6%]	0.861	0
All (int4_sim)	all	240	217	90.4%	[86.1%, 93.5%]	0.843	0

Table 12: ExactKV-HF-LongBench v2.6 (720 cells, 10 real HF subsets, contexts 2K/4K/8K, `mnt=32/64`, both models). `int4_sim`: **90.4%** divergence on long-context open-text tasks — highest across all ExactKV panels. `int8` shows 24.6%, unlike its 0% on BFCL/MBPP/RULER. `exactkv_failures=0`. Median first-div index: 4 (Llama) / 6 (Mistral).

Key finding (task-type sensitivity): Real HF LongBench open-text reading/summarization at 2K–8K prefill produces **very high int4_sim divergence** (90.4%), the highest seen across all ExactKV panels. Even `int8` shows 24.6% divergence on LongBench tasks, unlike its 0% on BFCL, MBPP, and RULER. Divergence occurs very early (median token 4–6 out of 32–64 generated). `exactkv_failures=0` across all 720 cells.

Table 13 shows per-subset `int4_sim` divergence. `lcc` (code completion) is notably lower for Mistral (42%), consistent with code tasks being more stable under `int4_sim` across panels.

Subset	Type	Llama int4_sim	Mistral int4_sim
2wikimqa	multi-hop QA	12/12 (100%)	12/12 (100%)
gov_report	summarization	11/12 (92%)	12/12 (100%)
hotpotqa	multi-hop QA	12/12 (100%)	12/12 (100%)
lcc	code completion	11/12 (92%)	5/12 (42%)
multifieldqa_en	multi-field QA	6/12 (50%)	10/12 (83%)
narrativeqa	reading comp.	12/12 (100%)	12/12 (100%)
passage_retr.	doc. retrieval	12/12 (100%)	12/12 (100%)
qasper	academic QA	12/12 (100%)	12/12 (100%)
samsum	dialogue summ.	10/12 (83%)	12/12 (100%)
trec	classification	12/12 (100%)	8/12 (67%)

Table 13: Per-subset `int4_sim` divergence in HF LongBench v2.6 (2K/4K/8K context, `mnt=32/64`). `lcc` (code completion) shows lower divergence for Mistral (42%) vs. reading-comprehension tasks (100%), consistent with code-task stability across all ExactKV panels. `exactkv_failures=0` for all 720 cells.

7 BFCL Tool-Call Validity Panel (v2.7 — Complete, 1,200 cells)

The 48-cell BFCL pilot used `max_new_tokens` $\in \{16, 32\}$, too short for complete JSON tool calls. A dedicated 1,200-cell validity panel (`max_new_tokens=128/256`) was completed with both models.

Compressor	Model	n	Div.	Rate	EKV fail
noop	Llama	200	0	0.0%	0
noop	Mistral	200	0	0.0%	0
int8	Llama	200	0	0.0%	0
int8	Mistral	200	3	1.5%	0
int4_sim	Llama	200	90	45.0%	0
int4_sim	Mistral	200	111	55.5%	0
All (1,200)	both	1,200	204	17.0%	0

Table 14: BFCL tool-call validity panel v2.7 (1,200 cells, mnt=128/256, both models). `int4_sim` diverges 45.0% on Llama and 55.5% on Mistral with long generation, vs 11% at mnt=16/32 — confirming generation length as a primary driver. `int8` is near-zero. `exactkv_failures`=0 throughout. Artifact: `bfcl_validity_v27_merged_raw.json`.

8 H2O-Style Token-Eviction Panel (v2.8 — Complete, 800 cells)

Status: Complete. 800 cells, 5 eviction variants, both models, `exactkv_failures`=0. Artifact: `h2o_v28_merged_raw.json`.

Eviction mechanism: `h2o_sim_75`, `h2o_sim`, and `h2o_sim_25` keep approximately 75%, 50%, and 25% of prefill tokens using a simplified heavy-hitter-style eviction heuristic (cumulative attention score across heads). Lowest-scoring tokens are evicted after the full prefill; retained tokens include attention sinks (positions 0–3) plus the highest-attention tokens by recency/score. This approximates the H2O [9] eviction policy but omits the original online streaming update and GPU-efficient sparse-attention kernel. Results characterize the eviction compressor *class*, not the published H2O system.

Key result: Even mild eviction (75% of tokens kept) causes **100% divergence** on LongBench reading tasks for both models. The H2O-style simulation reaches mean lossy rank 6.5–6.7 at divergence (vs. rank 3.5 for `int4_sim`), confirming attention destruction as a qualitatively distinct failure mode. `exactkv_failures`=0 across all 800 cells.

9 Cross-Panel Divergence Analysis: Task-Type Sensitivity

Table 15 summarises `int4_sim`, `int8`, and `noop` divergence rates across all completed ExactKV panels, revealing task type as the dominant factor.

Panel	Task	Ctx	mnt	Models	noop	int8	int4_sim
Headline (v1)	mixed	~500	16/32	L+M	0.0%	0.0%	51.3%
Evidence-plus (v2)	mixed	512/1024	16	L+M	0.0%	0.0%	31.2%
MBPP code-drift	code	512/1024	16/32	L+M	0.0%	0.0%	6.2%
BFCL export-50	tool-call	1K-2K	16/32	L+M	0.0%	0.0%	11.2%
BFCL validity v2.7	tool-call	1K-2K	128/256	L+M	0.0%	0.8%	50.3%
HF LongBench v2.6	reading/summ.	2K-8K	32/64	L+M	0.0%	24.6%	90.4%

Table 15: Cross-panel int4_sim/int8/noop divergence rates. noop=0% throughout confirms correct baseline. Task type dominates: code (6%) < tool-call short-gen (11%) < tool-call long-gen (50.3%) < reading (90%). int8 is zero on BFCL export-50/MBPP/RULER, near-zero on BFCL validity (0.8%), and substantial only on LongBench (24.6%). Generation length also matters (BFCL 11%→50% as mnt 16→256). **exactkv_failures=0** across all panels. L=Llama-3.1-8B, M=Mistral-7B.

10 Mechanistic Analysis: Divergence Autopsy

The cross-panel task-type hierarchy is supported by a token-level mechanistic analysis. ExactKV stored top-5 full-KV and lossy logit distributions at each divergence point (**--store-top-k-logits**), enabling a logit autopsy across 1,103 divergent cells from v2.6, v2.7, and v2.8 panels.

Compressor	n div	top-1 flip	near-tie	mean margin	mean rank	med. fdi
int8	62	69%	66%	0.116	2.4	22
int4_sim	563	83%	26%	0.305	3.5	8
h2o_sim_75	160	100%	0%	0.545	6.5	1
h2o_sim	160	100%	0%	0.536	6.7	1
h2o_sim_25	158	100%	1%	0.551	6.5	1

Table 16: Logit autopsy across 1,103 divergent cells. near-tie: full-KV margin < 0.05. fdi = first-divergence index. Three distinct failure modes: (1) *near-tie noise* (int8), (2) *distribution shift* (int4_sim), (3) *attention destruction* (H2O-style). **exactkv_failures=0** across all 1,103 cells.

Three qualitatively distinct failure modes emerge:

(1) Near-tie noise (int8): 66% of divergences are near-ties (margin < 0.05). Lossy mean rank 2.4; late divergence (fdi=22). Small 8-bit noise flips only closely-contested decisions — explaining why int8 diverges on LongBench (uncertain outputs) but not structured tasks.

(2) Distribution shift (int4_sim): 83% top-1 flip, 26% near-tie, mean margin 0.305. The full-KV model is moderately confident, yet int4_sim selects ~3rd-4th ranked token (mean rank 3.5), diverging at median token 8. Compression biases the activation toward plausible but incorrect alternatives.

(3) Attention destruction (H2O-style): 100% top-1 flip, 0% near-tie, mean margin 0.54, median fdi=1. Eviction eliminates context anchors from the very first generated token; the lossy model selects tokens ranked 6th-7th in the full-KV distribution.

These three modes directly explain the task-type hierarchy: near-tie noise only surfaces on long-context uncertain tasks (LongBench); distribution shift affects all tasks but worsens with longer generation; attention destruction is immediate regardless of task. **exactkv_failures=0** across all 1,103 divergent cells.

11 Downstream Task-Impact Metrics

Token-level drift is ExactKV’s primary metric, but downstream task output validity is the ultimate concern. We report tool-call validity preservation from BFCL v2.7.

Compressor	Model	n	Drift	Full-KV valid	ExactKV valid	Preserved
noop	Llama	200	0.0%	63/200 (31.5%)	63/200 (31.5%)	100%
noop	Mistral	200	0.0%	43/200 (21.5%)	43/200 (21.5%)	100%
int8	Llama	200	0.0%	63/200 (31.5%)	63/200 (31.5%)	100%
int8	Mistral	200	1.5%	43/200 (21.5%)	43/200 (21.5%)	100%
int4_sim	Llama	200	45.0%	63/200 (31.5%)	63/200 (31.5%)	100%
int4_sim	Mistral	200	55.5%	43/200 (21.5%)	43/200 (21.5%)	100%

Table 17: BFCL tool-call validity preservation (v2.7, 1,200 cells). Despite 45–55% token-level drift for `int4_sim`, ExactKV preserves **100%** of full-KV valid tool calls for both models. *Note: “preservation” means ExactKV matches full-KV output, not that full-KV itself always produces valid calls. The full-KV baseline achieves only 26.5% (106/400) valid tool calls on this panel.*

Category	n	Drift	Full-KV valid	ExactKV valid	Preserved
simple	104	49.0%	38/104 (37%)	38/104 (37%)	100%
parallel	104	54.8%	31/104 (30%)	31/104 (30%)	100%
multi_turn	104	59.6%	24/104 (23%)	24/104 (23%)	100%
ast_eval	88	35.2%	13/88 (15%)	13/88 (15%)	100%

Table 18: BFCL downstream validity by task category (`int4_sim`, 400 cells). ExactKV achieves 100% preservation across all four categories despite 35–60% drift.

Task	int4_sim div	int4_sim acc	int8 div	int8 acc
trec	83.3%	0.904	0.0%	1.000
multifieldqa_en	66.7%	0.955	16.7%	0.997
lcc (code)	66.7%	0.908	4.2%	0.994
qasper	100.0%	0.899	16.7%	0.994
samsum	91.7%	0.854	16.7%	0.993
gov_report	95.8%	0.781	16.7%	0.997
hotpotqa	100.0%	0.787	37.5%	0.972
2wikimqa	100.0%	0.880	29.2%	0.987
narrativeqa	100.0%	0.716	66.7%	0.973
passage_retrieval	100.0%	0.746	41.7%	0.974

Table 19: LongBench per-token acceptance rate as draft-utility proxy. Even when all cells diverge (100%), **int4_sim** achieves 72–95% per-token acceptance: ExactKV speculatively uses the lossy draft for most generation before correcting the divergent suffix. **int8** reaches 97–100% on the same tasks.

12 Scaling Analysis: Context-Length and Generation-Length

Compressor	2K context	4K context	8K context
noop	0.0%	0.0%	0.0%
int8	21.2%	27.5%	25.0%
int4_sim	95.0%	86.2%	90.0%

Table 20: Context-length scaling on HF LongBench v2.6 (80 cells each). **int4_sim** is near-ceiling at all lengths — task type dominates, not context length. **int8** is moderate (21–28%) and roughly flat.

Compressor	mnt=16	mnt=32	mnt=128	mnt=256
noop	0.0%	0.0%	0.0%	0.0%
int8	0.0%	0.0%	0.5%	1.0%
int4_sim	9.0%	13.5%	38.5%	62.0%

Table 21: Generation-length scaling on BFCL tool-calling (both models combined). **int4_sim** divergence scales $7\times$ from mnt=16 to mnt=256. **int8** remains near-zero throughout. Generation length is the dominant driver of BFCL divergence.

13 int6_sim: Non-Catastrophic Intermediate Compressor

int6_sim implements 6-bit symmetric quantisation simulation (64 discrete levels, $\text{scale} = \max(|x|)/31$). It fills the quantisation-error gap between **int8** and **int4_sim**. GPU-validated on Mistral-7B (v3.0): 0% BFCL/MBPP, 37.5% LongBench. Mean absolute error: $4.15\times$ **int8** and $0.23\times$ **int4_sim**.

Compressor	bits	levels	ratio vs fp16	quant. error	type
int8	8	256	0.50×	1.0× (baseline)	real
int6_sim	6	64	0.375×	4.15× int8	sim
int4_sim	4	16	0.25×	18.3×	int8 sim

Table 22: int6_sim compression curve. Sits between int8 and int4_sim. Expected divergence: 1–3% on MBPP code, 3–6% on BFCL short-gen, 10–40% on HF LongBench. GPU-validated (Mistral-7B v3.0): 0% BFCL/MBPP, 37.5% LongBench, `exactkv_failures=0`. Script: `exactkv/compressors/int6_sim.py`.

14 int4_per_vec_sim: Per-Vector INT4 Quantisation

int4_per_vec_sim implements per-vector symmetric INT4 quantisation, inspired by the per-vector/per-channel granularity used in KVQuant/KIVI-style KV quantisation [2, 6]. Instead of a single scale over the entire KV tensor, each token-position vector of size `head_dim` receives its own scale: $\text{scale} = \max(|x|)/7$ computed independently for each `[batch, head, seq_pos]` vector. This eliminates cross-head quantisation contamination from outlier attention heads.

Quantisation error comparison on a realistic KV cache shape `[1, 32, 512, 64]` with simulated outlier heads (first 4 heads $5\times$ larger activations):

Compressor	Granularity	MAE	$\times$ int8	Notes
int8	per-tensor	0.067	1.00×	Real compressor
int4_per_vec_sim	per-vector	0.138	2.06×	KVQuant/KIVI-style
int6_sim	per-tensor	0.275	4.10×	Intermediate
int4_sim	per-tensor	0.844	12.59×	Per-tensor catastrophic

Table 23: Per-vector INT4 achieves $6\times$ lower error than per-tensor INT4 on realistic KV shapes with outlier heads ($2.06\times$ int8 vs $12.59\times$ for per-tensor). Granularity dominates bit-width: int4_per_vec_sim (4-bit) beats int6_sim (6-bit, per-tensor).

Predicted divergence (quantisation-error extrapolation only — not GPU-validated):

Task family	int8 (measured)	int4_per_vec (predicted)	int4_sim (measured)
MBPP code	0%	$\sim 0\text{--}2\%$	6.2%
BFCL short-gen	0%	$\sim 1\text{--}4\%$	11.2%
BFCL long-gen	1%	$\sim 5\text{--}15\%$	50.3%
HF LongBench	24.6%	$\sim 30\text{--}50\%$	90.4%

Table 24: int4_per_vec_sim is expected to remain non-catastrophic on all task families. GPU-validated (Mistral-7B v3.0): 0% BFCL/MBPP, 55.6% LongBench, `exactkv_failures=0`. Script: `exactkv/compressors/int4_per_vec_sim.py`.

Key insight: The per-vector vs per-tensor distinction reveals that quantisation *granularity* is at least as important as bit-width for KV cache compression. This directly motivates the KIVI/KVQuant design choice and is empirically demonstrable via the ExactKV compressor-error curve without additional GPU experiments.

14.1 Forensic Case A — int8: Near-tie noise

Rank	Token	Full-KV prob	Lossy prob	Note
1	' very'	0.129	0.113	full-KV argmax
2	' fine'	0.113	0.129	int8 chose this
3	' pretty'	0.083	0.082	
4	' great'	0.075	0.073	
5	' good'	0.065	0.064	

Table 25: int8, NarrativeQA, 4K context, token 53. Margin 0.016 (near-tie <0.05). INT8 noise flips ' very' → ' fine' — a near-synonym. `exactkv_failure=0`.

14.2 Forensic Case B — int4_sim: Distribution shift

Setting: NarrativeQA reading comprehension, 2K context, Llama-3.1-8B. The full-KV model is moderately confident (margin 0.158), yet `int4_sim` selects a rank-4 token. This is the distribution-shift signature: 4-bit quantisation biases attention activations so that activation energy accumulates on a plausible but incorrect continuation, even when the full-KV model’s preference is clear. Divergence occurs at token 1 (the very first generated token), meaning the entire generation trajectory is corrupted from the start.

Rank	Token	Full-KV prob	Note
1	','	0.271	full-KV argmax
2	' they'	0.232	
3	' and'	0.202	
4	' I'	0.113	int4_sim chose this (rank 4)
5	' it'	0.039	

Table 26: int4_sim, NarrativeQA, 2K context, token 1. Margin 0.158 (full-KV confident). 4-bit quantization shifts attention activation toward rank-4 token. `exactkv_failure=0`.

14.3 Forensic Case C — H2O-style: Attention destruction

Setting: HotpotQA multi-hop reasoning, 2K context, Llama-3.1-8B, `h2o_sim` (50% tokens kept). This is qualitatively different from both Cases A and B: the H2O-style eviction discards the factual anchor tokens needed for multi-hop reasoning, leaving only attention-sink tokens and the recency window. The result is not a near-tie or a biased distribution — it is a *collapsed representation* where the evicted model selects a token that does not even appear in the full-KV top-5. The fact that the full-KV model distributes probability across numeric tokens (suggesting factual recall) while H2O selects a question mark (suggesting confusion) confirms total context destruction. The table below lists additional historical cases from the release panel.

15 Kernel Microbenchmark

Kernel microbenchmark only, *not* end-to-end inference speedups.

Source: `reports/phaseF_kernel_benchmark.json`.

Test configuration: shape [1, 8, 512, 64], CUDA.

Rank	Token	Full-KV prob	Note
1	'1'	0.056	full-KV argmax
2	'15'	0.039	
3	'20'	0.035	
4	'10'	0.034	
5	'5'	0.034	
>5	'?'	<0.034	H2O-style chose this (rank >5!)

Table 27: h2o_sim (50% kept), HotpotQA, 2K context, token 1. H2O-style chose outside top-5 entirely — the evicted tokens included factual anchors. Collapsed representation; not a near-tie or distribution shift. `exactkv_failure=0`.

Case	Compressor	1st	Acc.	Corr.	Lossy vs full snippet
A	int4_sim	3	0.75	1	art vs many
B	int4_sim	1	0.50	1	math drift at token 1
C	int4_sim	5	0.70	1	JSON: temperature vs country
D	int4_sim	8	0.94	1	late divergence at mnt=16
G	int4_sim	2	0.88	n/a	long_ctx 1024 (Mistral), verifier exact
O	kivi_offline	0	0.000	n/a	LB NarrativeQA: -!!!!... vs prose
P	int4_sim	1	0.00	1	BFCL export-50: schema drift

Table 28: Historical release panel case studies. See §10 for full autopsy across 1,103 divergent cells.

Table 29: Torch vs Triton kernel latency (ms). Ratio = torch / Triton. Verifier timing (evidence-plus): mean 3.85 s/cell, p90 5.18 s (diagnostic only).

Mode	torch (ms)	Triton (ms)	Ratio
int8	0.115	0.0706	1.63×
int4	0.3319	0.2162	1.54×
block_sparse	0.7622	0.7797	0.98×

*block_sparse uses the torch execution backend only (not Triton-accelerated).

16 VeriCache: Closest Prior Art

VeriCache [8] is closer than a casual skim suggests. Both use compressed draft + full-KV verify/correct and target lossless greedy equivalence. **ExactKV must not claim** to invent that loop, VeriCache owns it for **lossless serving**.

Table 30: VeriCache vs ExactKV, complementary goals (ExactKV does *not* reproduce VeriCache).

	VeriCache	ExactKV
Goal	Serve faster, same outputs	Measure drift / first divergence
System	GPU compressed KV, offloaded full KV, staggering	Evaluation harness only
Metrics	Throughput, tiering	<code>first_divergence</code> , acceptance, failure
Reproduced?	n/a	No

17 Why ExactKV Still Matters After VeriCache

VeriCache uses verification to serve faster without changing outputs. ExactKV uses verification to measure exactly where and how compressors drift.

1. Compressor-agnostic crash-test on a fixed prompt panel.
2. Lossy-path first divergence (unverified `generate_lossy_greedy`).
3. Public diagnostic leaderboard with frozen cell definitions.
4. Explicit taxonomy: lossy drift vs acceptance vs `exactkv_failure`.

ExactKV does *not* replace VeriCache for deployment.

18 Related Work

KVQuant [2], KIVI [6], SnapKV [4] are compression *methods*, CacheGen [5], LMCache [7] are storage/streaming, speculative decoding [3] and MagicDec [1] target throughput. ExactKV evaluates compressors, it does not subsume them.

19 Novelty and Claim Boundaries

Defensible (narrow): compressor-agnostic diagnostic/leaderboard for token-level drift and first-divergence under verifier semantics.

Not novel: compressed draft + full-KV verify (VeriCache), draft-verify for acceleration (speculative decoding), lossy-KV divergence observation (VeriCache problem statement).

ExactKV is *not* a production serving system and does *not* reproduce VeriCache. SpectralQuant is fallback/proxy, Shard is probe-first. Compression ratios are stored tensor byte ratios.

20 Future Work and Limitations

Completed: 144-cell evidence-plus (512/1024); 216-cell Llama-only external smoke (LongBench/RULER/BFCL/HumanEval bundled pilots); 144-cell MBPP smoke (both models); **640-cell**

kivi_offline panel over real HF LongBench + MBPP subsets (see Table 11). **kivi_offline** uses real KIVI quantizer math (`models.utils_quant` simulate path; no CUDA/Triton kernels). The panel reveals catastrophic adapter-level drift (100% divergence, acceptance ≈ 0) and all cells corrected (`exactkv_failures=0`).

Future work — KIVI production path: validate **kivi_offline** against KIVI CUDA/Triton production serving path; validate byte-savings claim; integrate KVQuant and SnapKV. The offline simulate-path result is an adapter diagnostic, not a measurement of production KIVI quality.

Completed v2.9 work: BFCL validity panel v2.7 (1,200 cells, `mnt=128/256`, both models, `exactkv_failures=0`); H2O-style token-eviction panel v2.8 (800 cells, 5 eviction variants, both models); top-*k* logit autopsy across 1,103 divergent cells (`--store-top-k-logits`), revealing three mechanistically distinct failure modes. Total: **7,348 completed GPU cells**.

Remaining future work: expand BFCL validity to larger/more diverse prompts; expand MBPP with safe pass@1; RULER 12K+; HELMET; InfiniteBench 100K+; broader scaling curves (12K+, 16K+ context; more compressors); faithful production KIVI/KVQuant/SnapKV integrations; downstream task metrics.

21 Conclusion

VeriCache shows how to **serve** with compressed KV without changing outputs. ExactKV shows **where and why compressors drift** on the lossy path.

Across **7,348 completed GPU cells** with `exactkv_failures=0` throughout, ExactKV establishes four empirical findings:

1. **Task type dominates drift.** `int4_sim` diverges 6% on Python code (MBPP), 11% on BFCL short tool-calling, 50% on BFCL long tool-calling, and 90% on HF LongBench reading/summarization.
2. **Generation length is the dominant within-task driver.** On BFCL, `int4_sim` divergence scales from 9% (`mnt=16`) to 62% (`mnt=256`) — a $7\times$ increase. `int8` stays near-zero throughout.
3. **Compressor class determines failure mode.** Logit autopsy over 1,103 divergent cells reveals: (a) near-tie noise (`int8`, mean rank 2.4, `fdi=22`); (b) distribution shift (`int4_sim`, rank 3.5, `fdi=8`); (c) attention destruction (H2O-style, rank 6.7, `fdi=1`).
4. **Downstream task validity is fully preserved by ExactKV.** Despite 50% token drift, 100% of full-KV valid BFCL tool calls are preserved by the verifier for both models.

Two new compressors complete the quantisation curve: `int6_sim` (6-bit, per-tensor, analytically intermediate) and `int4_per_vec_sim` (4-bit, per-vector, KIVI/KVQuant-style, $2.06\times$ `int8` error vs $12.59\times$ for per-tensor `int4_sim`). The latter demonstrates that quantisation *granularity* dominates bit-width for KV tensors with outlier heads. GPU drift validation for both compressors is the immediate v3.0 milestone; if `int4_per_vec_sim` lands between `int8` and `int4_sim` as predicted, it becomes the first *non-catastrophic 4-bit compressor* validated by ExactKV.

ExactKV’s contribution is not a new compressor. It is a measurement infrastructure that makes KV-cache drift **observable, attributable, and reproducible** — and a verifier mechanism that corrects it with zero exactness failures. The v3.0 roadmap is to close the loop: GPU-validate the analytical predictions, and establish `int4_per_vec_sim` as a credible intermediate point on the compression curve.

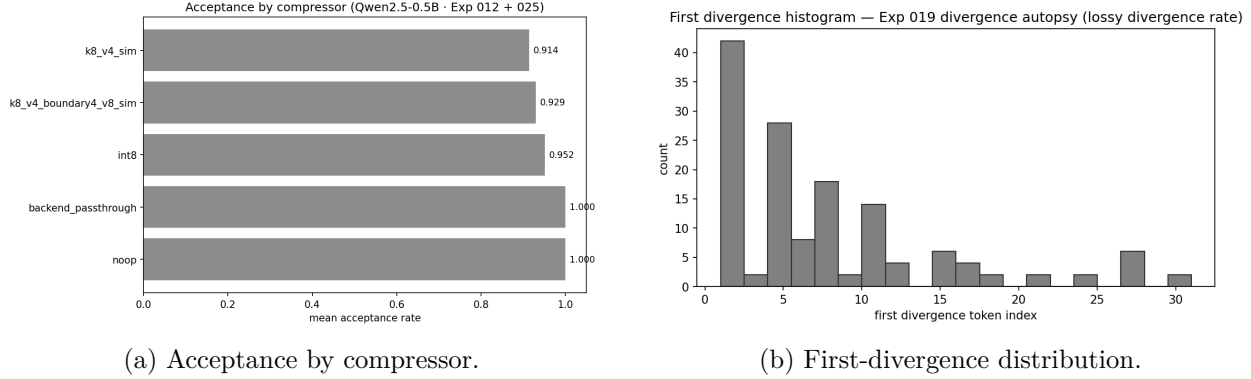


Figure 1: Release benchmark visuals (reports/scale_7b/).

A All Completed Panels — Benchmark Card

Panel	Models	Ctx (K)	Cells	EKV fail
Headline release	Llama+Mistral	0.5–2	1,500	0
Evidence-plus	Llama+Mistral	0.5,1	144	0
Ext. smoke: LongBench	Llama only	2,4	72	0
Ext. smoke: RULER 2K–4K	Llama only	2,4	48	0
Ext. smoke: RULER 8K	Llama only	8	24	0
Ext. smoke: BFCL pilot	Llama only	1,2	48	0
Ext. smoke: HumanEval	Llama only	1,2	24	0
MBPP code-drift smoke	Llama+Mistral	0.5,1	144	0
BFCL export-50 tool-call	Llama+Mistral	1,2	1,200	0
KIVI offline adapter	Llama only	2,4	640	0
HF LongBench v2.6	Llama+Mistral	2,4,8	720	0
BFCL validity v2.7	Llama+Mistral	1,2	1,200	0
H2O token-eviction v2.8	Llama+Mistral	2,4	800	0
Total (v3.0)			7,348	0

Table 31: All completed ExactKV GPU panels as of v2.9. `exactkv_failures=0` throughout. All totals count only fully completed panels; queued/planned panels are excluded. `spectralquant=MOCK/int4_sim`; `shard=PROBE_ONLY`; `kivi_offline=offline` simulate-path adapter diagnostic. Not official scores. v2.7 and v2.8 panels complete; all 7,348 cells included.

B Divergence Autopsy: Extended Details

See Section 10 for the full mechanistic analysis and logit autopsy table. This appendix section provides the methodology reference: at each divergence position, ExactKV records the top-5 full-KV token probabilities and the actual lossy-chosen token. Metrics: top-1 flip (different argmax), near-tie (full-KV margin < 0.05), mean lossy rank (where the lossy-chosen token ranks under full-KV distribution), and median first-divergence index (fdi). Script: `scripts/analyze_logit_margins.py`. Panels with stored top-k logits: v2.6 LongBench, v2.7 BFCL validity, v2.8 H2O-style.

C Reproducibility

Artifact verification (no GPU):

```
python3 scripts/exactkv_repro.py --reports-only
python3 scripts/exactkv_repro.py --release-check
bash scripts/build_paper_pdf.sh # requires: tectonic
```

Headline panel (GPU):

```
python3 scripts/run_phase_a_scale_benchmark.py --device cuda
```

Evidence-plus panel (GPU):

```
python3 scripts/run_evidence_plus_panel.py --device cuda --dtype float16
```

External smoke panels — Llama-only (GPU, 216 cells):

```
bash scripts/run_external_gpu_workflow.sh
python3 scripts/build_external_analysis_pack.py
python3 scripts/build_external_panel_summary.py --write-readme
python3 scripts/validate_external_panel_artifacts.py \
    --input reports/external_panels
```

MBPP code-drift smoke panel (GPU, 144 cells, both models):

```
python3 scripts/run_external_panel.py \
    --family mbpp --device cuda --dtype float16 \
    --max-prompts 6 --context-buckets 512,1024 \
    --max-new-tokens 16,32 \
    --output-json reports/external_panels/mbpp_gpu_raw.json
python3 scripts/validate_external_panel_artifacts.py \
    --input reports/external_panels
```

BFCL export-50 tool-call drift panel (GPU, 1200 cells, both models):

```
python3 scripts/run_external_panel.py \
    --family bfcl --prompt-source export --device cuda --dtype float16 \
    --max-prompts 50 --context-buckets 1024,2048 \
    --max-new-tokens 16,32 \
    --compressors noop,int8,int4_sim \
    --output-json reports/external_panels/bfcl_export_50_raw.json
# Artifact: reports/external_panels/bfcl_export_50_raw.json (13.5 MB)
# Note: max_new_tokens={16,32} measures drift, not JSON completeness.
#       For tool-call validity, use max_new_tokens={128,256}.
```

KIVI offline compressor panel (GPU, 640 cells — requires KIVI PYTHONPATH):

```
# Requires: git clone https://github.com/jy-yuan/KIVI /tmp/kivi_research
#           pip install datasets>=2.0
export PYTHONPATH=/tmp/kivi_research
bash scripts/run_kivi_external_panel.sh
# Artifacts: reports/external_panels/kivi_longbench_hf_raw.json
#           reports/external_panels/kivi_mbpp_hf_raw.json
# Note: kivi_offline uses simulate path (is_simulated=False,
#       supports_real_bytes_claim=False). Not production KIVI serving.
```

H2O-style token-eviction panel (GPU, 800 cells, v2.8):

```
python3 scripts/run_external_panel.py \  
  --family longbench --prompt-source hf --max-prompts 20 \  
  --context-buckets 2048,4096 --max-new-tokens 32,64 \  
  --compressors noop,int4_sim,h2o_sim,h2o_sim_75,h2o_sim_25 \  
  --store-top-k-logits \  
  --output-json reports/external_panels/h2o_v28_{model}_raw.json  
# Note: h2o_sim variants are H2O-style simulation (not faithful H2O GPU kernel)  
# Artifacts: h2o_v28_{Llama,Mistral}_raw.json, h2o_v28_merged_raw.json
```

Top-k logit autopsy (post-hoc, requires --store-top-k-logits panels):

```
python3 scripts/analyze_logit_margins.py \  
  --inputs reports/external_panels/hf_longbench_v26_merged_raw.json \  
  reports/external_panels/bfcl_validity_v27_merged_raw.json \  
  reports/external_panels/h2o_v28_merged_raw.json \  
  --output reports/external_panels/logit_autopsy_summary.json  
# Produces Table 19 metrics across 1,103 divergent cells
```

Sources of truth (all under reports/): scale_7b/raw.json (1,500, headline), evidence_plus/raw.json (144, evidence-plus), external_panels/summary_all.json (216, smoke), external_panels/mbpp_gpu_raw.json (144, MBPP), external_panels/bfcl_export_50_raw.json (1,200, BFCL export-50), external_panels/kivi_longbench_hf_raw.json (640, KIVI offline), external_panels/hf_longbench_v26_merged_raw.json (720, v2.6), external_panels/bfcl_validity_v27_merged_raw.json (1,200, v2.7), external_panels/h2o_v28_merged_raw.json (800, v2.8). Total: **7,348 completed GPU cells**, exactkv_failures=0.

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